Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/27/2008
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Project Title:	Effects of Simulated Weightlessness on the Repair of Lower Limb Bone Fractures and on the Number of Bone-derived Stem Cells		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44195	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Muschler, George (The Cleveland Clinic Foundation) Vasanji, Amit (The Cleveland Clinic Foundation) Cavanagh, Peter (University of Washington)		
Grant/Contract No.:	NCC 9-58-BL00405		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Aim 1: Assess the rate and extent of fracture healing in osteotomized rats in simulated weightlessness versus normal gravity. Aim 2: Assess the numbers of progenitor cell colony-forming units (CFU) and osteogenic cells per CFU in bone marrow and periosteum tissues from osteotomized rats in simulated weightlessness versus normal gravity. Aim 3: Determine the rate and extent of fracture healing in osteotomized rats in simulated weightlessness when treated with PTH or bisphosphonate therapy. Aim 4: Assess the numbers of progenitor cell CFU and osteogenic cells per CFU in bone marrow and periosteal tissues from osteotomized rats in simulated weightlessness undergoing either PTH or bisphosphonate therapy. In year 1, rats experiencing normal gravity (weight bearing or WB) exhibited a spontaneous repair of cortical bone trauma within a 5-week healing period. Intermittent PTH therapy enhanced this WB repair process by: 1) increasing the rate of hard callus formation and peak hard callus volume compared to vehicle-control rats; 2) increasing bone formation rates (BFR); and 3) increasing the number of osteoprogenitors within bone marrow tissue. In year 2, rats experiencing hind limb unloading (non-weight bearing or NWB) exhibited a greatly diminished level of spontaneous repair of bone trauma as compared to the WB group within the same healing period. This was reflected in a diminished maximum hard callus volume and a rate of hard callus formation that were both only ~20% of those of the WB group. Reacquisition of a weight-bearing status equal in time to that of the period of simulated weightlessness prior to bone trauma (NWB-WB group) did not substantially improve the bone healing response. In this case, this group exhibited a maximum hard callus volume that was only 34% of, and a rate of hard callus formation that was only 28% of the WB group. Both NWB and NWB-WB groups exhibited low bending strength values at their respective callus sites by 5-weeks post-op (0.28 ± 0.05 N/m) in stark contrast to the bending strength of the WB callus (1.75 ± 0.35 N/m). Histology revealed that 46% of the NWB group fibulae were reconnected by a fibrous soft tissue union (“non-union”), while all of the WB group specimens were reconnected by a bony tissue union. Lastly, NWB reduced the number of marrow progenitor cells and osteoprogenitors by 90% from the levels exhibited by WB bone specimens. In year 3, intermittent PTH therapy (80 ug/kg BW) was shown to partially reverse the deleterious effects of NWB on cortical bone trauma healing. Only 12.5% of NWB-PTH specimens were classified as “non-unions” (a 73% decrease from that of NWB) and 50-75% increases in maximum hard callus volume and hard callus formation rates were observed as compared to NWB specimens. PTH therapy increased BFR by ~2-fold over those exhibited by NWB rats. PTH therapy enhanced the numbers of osteoprogenitor cells by ~4-fold as compared to those measured from NWB specimens. We are currently assessing bending strength and histology of these specimens. Altogether, intermittent PTH therapy partially restores select aspects of bone healing under simulated microgravity. Grant BL00405 was extended for an extra 6 months of work (04/01/07 – 09/01/07) and during this time NWB rats were treated with alendronate (bisphosphonate). Currently, the results from this trial are under analysis and results will be forth coming. Findings from BL00405 indicate that cortical bone fracture healing is impaired in hind limb unloaded rats. The implication of these findings is that bone trauma repair in astronauts on long space missions would likely be compromised, and presents a potential threat to mission effectiveness and astronaut health. Further, these findings also suggest that astronauts who return from extended spaceflight missions are at continued risk for impaired bone trauma repair after returning to a normal gravity environment even up to a period of time equal to that of the space mission. Secondly, findings from BL00405 indicate that extended exposure to hind limb unloading results in drastic reductions in both total marrow progenitor cells and osteoprogenitors. Thus, a simulated microgravity situation appears to cause a reduction in the number of preosteoblasts that would be needed to generate a functional osteoblast population needed to repair bone trauma. These findings provide a plausible mechanism of action underlying this impaired fracture healing response. Thirdly, findings from BL00405 indicate that intermittent PTH therapy improves fracture healing under chronic hind limb unloading, though it does not normalize fracture healing to WB levels. The dramatic reduction in the number of fracture “non-unions” resulting from PTH therapy suggests that this treatment is a viable candidate to be tested as a countermeasure to offset any potential deleterious effects of weightlessness on bone trauma healing. Fourthly, findings from BL00405 indicate that PTH therapy significantly increased the number of functional osteogenic progenitor cells in bone under chronic hind limb unloading. PTH therapy’s increase in the number of osteoprogenitors correlated with its improvements in fracture healing in NWB rats strongly suggests a cause and effect between these parameters and strengthens our hypothesis that a decrease in progenitor cells represents a mechanism of action underlying the impairment to fracture healing. If correct, then the implications of these findings are that marrow progenitor cell populations may be altered in astronauts on long space missions and might manifest in deficiencies in musculoskeletal tissue repair after trauma. Given that BL00405 has reached is closure date, our remaining plans are to complete the remaining data assessments and submit manuscripts for submission to peer-reviewed publications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impact of our findings resulting from BL00405 suggest that impaired bone healing during and after an extended exposure to simulated weightlessness should manifest relevance to the underlying causes of impaired bone healing in patients experiencing paralysis, chronic immobility or extended bed rest. Our data indicating that intermittent PTH therapy can counteract the impairment of bone healing under a non-weight bearing situation may offer a potential treatment for augmenting bone healing in these Earth-bound, non-weight bearing patients. Our studies found that PTH therapy significantly increased the prevalence of osteogenic progenitor cells within bone marrow tissue over that measured in vehicle-control rats (normal gravity group). Further, PTH therapy increased the prevalence of osteogenic progenitor cells within bone marrow tissue 4-fold over that measured in vehicle-control hind limb suspended rats (chronic non-weight bearing). Coupled with the observations that PTH therapy improved fracture healing in hind limb suspended rats, these findings suggest that an underlying mechanism of action of PTH in its anabolic activity for enhancing bone fracture repair lies at the level of increasing the pool size of functional bone cells derived from osteogenic precursor cells. These same findings suggest that PTH therapy should aid in the healing of recalcitrant bone fractures here on Earth particularly in patients experiencing paralysis, chronic immobility or extended bed rest.

Task Progress:	The task progress of BL00405 directly addresses aspects of Critical Risk #2: Impaired Fracture Healing (Bone fractures incurred during and immediately after long duration space flight may require a prolonged period for healing, and the bone may be incompletely restored due to changes in bone metabolism associated with space flight). (1) task progress shows that prolonged exposure to hind limb unloading resulted in impaired fibular fracture healing. This directly addresses CPR #2.11: Does prolonged exposure to hypogravity lead to impaired healing of fractures? Our work raises the CRL of this enabling question from an original 1-2 to a 3-4 level. (2) task progress shows that prolonged exposure to hind limb unloading decreased the population size of bone marrow osteoprogenitor cells. This directly addresses CPR #2.23: Does hypogravity affect the size, viability, or differentiation of precursor bone cell populations? Our work raises the CRL of this enabling question from an original 1 to a 2-3 level. (3) task progress shows that prolonged exposure to hind limb unloading decreased the osteogenic differentiation of bone marrow osteoprogenitor cells. This directly addresses CPR #2.23: Does hypogravity affect the size, viability, or differentiation of precursor bone cell populations? Our work raises the CRL of this enabling question from an original 1 to a 2-3 level. (4) task progress shows that PTH therapy improves bone healing outcomes incurred by prolonged exposure to hind limb unloading (CPR #2.06: What pharmacological agents will most effectively minimize the decrease in bone mass with hypogravity? Are anabolic as well as anti-resorptive agents required?). Our work raises the CRL of this enabling question from an original 1 to a 2-3 level. (5) task progress shows that PTH therapy enhances bone formation rates in rats exposed to chronic hind limb unloading thereby boosting the rate of bone production under simulated weightlessness (CPR #2.06: What pharmacological agents will most effectively minimize the decrease in bone mass with hypogravity? Are anabolic as well as anti-resorptive agents required?). Our work raises the CRL of this enabling question from an original 1 to a 2-3 level. (6) task progress shows that PTH therapy improves the population size of marrow osteoprogenitors in rats exposed to chronic hind limb unloading (CPR #2.06: What pharmacological agents will most effectively minimize the decrease in bone mass with hypogravity? Are anabolic as well as anti-resorptive agents required?). Our work raises the CRL of this enabling question from an original 1 to a 2-3 level. Altogether, our findings suggest that fracture healing may be adversely affected by chronic exposure to weightlessness, and PTH injections, as an anabolic bone therapeutic approach, may be used to ameliorate some of these adverse effects on bone trauma healing.
Bibliography Type:	Description: (Last Updated: 03/01/2017)
Articles in Other Journals or Periodicals	Midura RJ, Su X, Androjna C. "Nonweight-Bearing state diminishes cortical bone healing response." Cleveland Clinic Orthopaedic Insights, Fall 2006. , Sep-2006
Articles in Peer-reviewed Journals	Midura RJ, Su X, Androjna C. "A simulated weightlessness state diminishes cortical bone healing responses." J Musculoskelet Neuronal Interact. 2006 Oct-Dec;6(4):327-8. PMID: 17185809 , Oct-2006
Books/Book Chapters	Androjna C, Su X, Cavanagh PR, Midura RJ. "Fracture healing may be impaired during spaceflight." in "Bone loss during spaceflight : etiology, countermeasures, and implications for bone health on Earth." Ed. P.R. Cavanagh, A.J. Rice. Cleveland : Cleveland Clinic Press, c2007, p. 93-101., Dec-2007
Papers from Meeting Proceedings	Androjna C, Su X, Midura RJ. "A simulated weightlessness state diminishes cortical bone healing responses." 36th International Sun Valley Workshop on Skeletal Tissue Biology, Sun Valley, ID, July 30 - Aug 3, 2006. Summaries and Abstracts, 36th International Sun Valley Workshop on Skeletal Tissue Biology, Sun Valley, ID, July 30 - Aug 3, 2006. p. 18-20. , Aug-2006
Significant Media Coverage	Midura RJ, Cavanagh PR. "NCRM Cleveland Press Release: National Center for Regenerative Medicine researchers find connection between chronic limb immobility and inhibited bone regeneration." NCRM Cleveland Press Release, May 2007., May-2007